

Lorentz Currents

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1 Lorentz Currents

A global lorentz transformation $x^\mu \rightarrow \Lambda^\mu{}_\nu x^\nu$ should produce a conserved current K . Now we try to derive such a current.

Consider a infinitesimal lorentz transformation

$$\Lambda^\mu{}_\nu = I^\mu{}_\nu + \frac{1}{2}\omega_{\alpha\beta}J^{\alpha\beta\mu}{}_\nu + \dots$$

where J is antisymmetric in the first two indices :

$$J^{\alpha\beta\mu}{}_\nu = -J^{\beta\alpha\mu}{}_\nu$$

then δx^μ is the difference between such a small spin and the original x^μ , that is

$$\delta x^\mu = \frac{1}{2}\omega_{\alpha\beta}J^{\alpha\beta\mu}{}_\nu x^\nu.$$

Now we turn to look at the arbitrary current J^μ . Such a current must satisfy $\partial_\mu J^\mu = 0$, which lead to

$$\partial_\mu \left[\mathcal{L} \delta x^\mu + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta \phi \right] = 0$$

it is then obvious that

$$\delta \phi = \phi(x^\mu) - \phi(\Lambda^\mu{}_\nu x^\nu) = -\frac{1}{2}\omega_{\alpha\beta}J^{\alpha\beta\mu}{}_\nu x^\nu \partial_\mu \phi$$

then the whole equation becomes

$$\partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\lambda \phi \omega_{\alpha\beta} J^{\alpha\beta\lambda}{}_\nu x^\nu - \mathcal{L} \omega_{\alpha\beta} J^{\alpha\beta\mu}{}_\nu x^\nu \right] = 0$$

consider an antisymmetric tensor $g_{\mu\nu}$. We will have the following:

$$\begin{aligned} g_{\mu\nu} x^\mu x^\nu &= \left[\frac{1}{2}(g_{\mu\nu} + g_{\nu\mu}) + \frac{1}{2}(g_{\mu\nu} - g_{\nu\mu}) \right] x^\mu x^\nu \\ &= \frac{1}{2}(g_{\mu\nu} x^\mu x^\nu - g_{\nu\mu} x^\mu x^\nu) \\ &= \frac{1}{2}g_{\mu\nu}(x^\mu x^\nu - x^\nu x^\mu). \end{aligned}$$

Then the whole current-conservation-equation will become

$$\begin{aligned}
\partial_\mu \omega_{\nu\lambda} \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\lambda \phi x^\nu - \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\nu \phi x^\lambda - \mathcal{L} g^{\lambda\mu} x^\nu + \mathcal{L} g^{\nu\mu} x^\lambda \right] &= 0 \\
\partial_\mu \omega_{\nu\lambda} \left[\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\lambda \phi - \mathcal{L} g^{\lambda\mu} \right) x^\nu - \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\nu \phi - \mathcal{L} g^{\nu\mu} \right) x^\lambda \right] &= 0 \\
\partial_\mu \omega_{\nu\lambda} \left[T^{\mu\lambda} x^\nu - T^{\mu\nu} x^\lambda \right] &= 0 \\
\omega_{\nu\lambda} \left(T^{\mu\lambda} x^\nu - T^{\mu\nu} x^\lambda \right) &= J^\mu
\end{aligned}$$

in which we used the fact that

$$J^{\alpha\beta\mu}_{\nu} = J^{\alpha\beta}_{\lambda\nu} g^{\mu\lambda} = (\delta^\alpha_\lambda \delta^\beta_\nu - \delta^\alpha_\nu \delta^\beta_\lambda) g^{\mu\lambda}.$$

Having ω as an arbitrary parameter, the part inside the brackets is a conserved tensor as well, namely

$$\partial_\mu \left(T^{\mu\lambda} x^\nu - T^{\mu\nu} x^\lambda \right) = 0 \implies \partial_\mu K^{\mu\nu\lambda} = 0$$

where the $K^{\mu\nu\lambda}$ is our Lorentz Current.